

Machine Learning Models

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Abstract

This document summarizes key equations and loss functions for popular machine learning models.
Notation: x = input, y = target, $\hat{y}$ = prediction, w = weights, λ = regularization strength.

Contents

I. Supervised Learning

1. Linear Models

- **Linear Regression:**

$$\hat{y} = w^T x + b$$
$$\mathcal{L}(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (\text{MSE})$$

- **Logistic Regression:**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$
$$\hat{y} = \sigma(w^T x + b)$$
$$\mathcal{L}(w, b) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (\text{Cross-Entropy})$$

- **Ridge/Lasso:**

$$\mathcal{L}_{\text{Ridge}} = \text{MSE} + \lambda \|w\|_2^2$$
$$\mathcal{L}_{\text{Lasso}} = \text{MSE} + \lambda \|w\|_1$$

2. Non-Linear Models

- **Decision Tree (Classification):**

$$\text{Gini} = 1 - \sum_{k=1}^K p_k^2, \quad \text{Entropy} = - \sum_{k=1}^K p_k \log p_k$$

- **Random Forest:**

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (\text{Ensemble averaging})$$

- **Gradient Boosting:**

$$\hat{y}^{(m)} = \hat{y}^{(m-1)} + \eta \cdot f_m(x)$$
$$f_m = \arg \min_f \sum_{i=1}^n L(y_i, \hat{y}^{(m-1)} + f(x_i))$$

II. Unsupervised Learning

1. Clustering

- K-Means:

$$\min_{\{C_k\}} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

2. Dimensionality Reduction

- PCA:

$$W^* = \arg \max_W \text{Var}(XW) \quad \text{s.t.} \quad W^T W = I$$

III. Deep Learning

1. Neural Networks

- Forward Pass:

$$a^{(l)} = f(W^{(l)} a^{(l-1)} + b^{(l)}) \quad (\text{e.g., ReLU: } f(z) = \max(0, z))$$

2. CNNs

- 2D Convolution:

$$(I * K)_{ij} = \sum_m \sum_n I(i+m, j+n) K(m, n)$$

IV. Reinforcement Learning

1. Q-Learning

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

2. Policy Gradient

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a)]$$

References

- Hastie et al. (2009) *The Elements of Statistical Learning*
- Goodfellow et al. (2016) *Deep Learning*